

STA130 Project

Project subtitle (if applicable)

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Overall Theme: [Factors associated with electricity access and performance in African countries.]

Research Goal: [The research goal is to examine whether GDP per capita is a significant predictor of electricity access rates across African countries.]

Target Population: [Yifan , Lucas , Natalia , Linus]

Audience: [Who is this for?]

Key Variables:

- Variable 1: [GDP per capita (measure of economic development for each country)]
- Variable 2: [Electricity access rate (percentage of population with access to electricity)]

Data Wrangling:

- [Aggregated the dataset to one observation per country by taking averages
Removed missing values for GDP and electricity access rate
Created GDP groups (high vs low) based on the median GDP per capita]

Question #3: [Lucas - What is the distribution of electricity outage duration (SAIDI) across African countries?]

This question examines how electricity outage duration varies across African countries. SAIDI measures the total outage time experienced over a year, so it is a direct indicator of electricity reliability. Countries with higher SAIDI values experience longer total outage duration, which signals weaker electricity system performance.

Q3: Variables & Approach

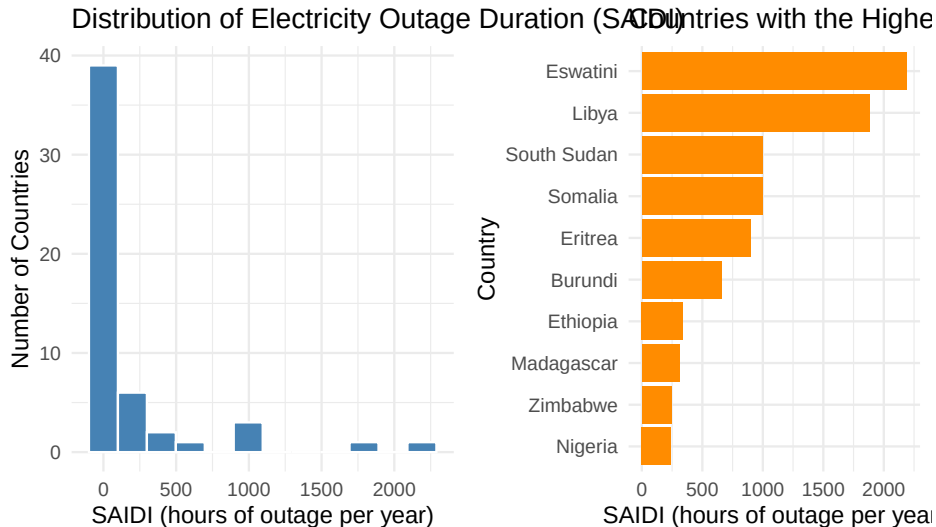
Variables used:

- country
- SAIDI

Approach:

- Use the cleaned country-level dataset CATF_clean
- Remove countries with missing SAIDI values
- Show the overall distribution of SAIDI using a histogram
- Show the countries with the highest SAIDI values using a bar chart

Q3: Visualization



Q3: Summary Table

Table 3: Summary of SAIDI Across African Countries

Countries	Median SAIDI	Mean SAIDI	IQR SAIDI	Minimum SAIDI	Maximum SAIDI
53	41.7	194.1	91.3	0	2190

Insights for Q3

- SAIDI varies substantially across African countries, showing that electricity reliability is uneven across the region.
- The histogram shows whether most countries have relatively low outage duration or whether the distribution is pulled upward by a smaller number of high-outage countries.
- The bar chart identifies the countries with the highest outage burden, which helps highlight where reliability problems appear most severe.

Overall Limitations

It is optional to include this as a stand-alone section at the end, you could also choose to discuss this at the end of each research question

Data Limitations:

- [The dataset contains missing values for some countries, which may reduce the representativeness of the analysis]
- [The data is aggregated at the country level, which may hide variation within countries (e.g., urban vs rural differences)]

Future Directions:

- [Future analysis could include additional variables such as infrastructure development, policy factors, and geographic conditions to better explain differences in electricity access]

Overall Conclusion

Key Findings:

- [GDP per capita is positively associated with electricity access across African countries]
- [Countries with higher GDP per capita have significantly higher electricity access rates]
- [High-GDP countries have about 26 percentage points higher electricity access than low-GDP countries]

Connection to Theme:

[The findings highlight the importance of economic development in improving electricity access, which is a key component of energy sustainability and infrastructure development across African countries]